Tshwane University of Technology

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INSTRUCTIONS TO CANDIDATES

1. All examination rules stated by the Tshwane University of Technology apply.
2. Ensure a single final version of your source code is handed in as requested.
3. State all necessary assumptions clearly in code commentary where needed.

MARKS: 50

PAGES: 11 (incl. cover)

EXAMINER:

Mr A.J. Smith

Mr M. Chauke

Prof J.A. Jordaan

MODERATOR:

Mr D Engelbrecht

TIME:

90 Minutes

15 Minutes extra time in the event of computer problems

FACULTY OF ENGINEERING AND THE BUILT ENVIRONMENT

DEPARTMENT OF ELECTRICAL ENGINEERING

ES216BB ENGINEERING SOFTWARE DESIGN B

EVALUATION 1 D

AUGUST 2026

EVALUATION INSTRUCTIONS

1. **Plagiarism:** Submit only original work. Similarity software will verify the authenticity of all submissions.

2. **Permitted Tools:** Use only CodeBlocks and Google Chrome to access the evaluation, view the paper, and upload your submission. Email, other online resources, memory sticks and unauthorised tools are prohibited.

3. **File Submission:** Rename the supplied source template to <student number>.cpp. Do not add a name, surname or other text to the file name.

4. **Uploading Instructions:** ONLY SUBMIT YOUR SOURCE FILE (.cpp) through the designated upload link. The latest submission is retained.

5. **Evaluation Scope:** This assessment covers Engineering Software Design B Units 1 to 3.

6. **Programming Language:** Construct the required program in C++11 and follow structured programming principles.

7. **Editing and Requirements:** Meet all specified requirements and use the attached appendices as required.

8. **Evaluation Requirements:**

    a. Remember to save your work on the PC D: drive and save regularly throughout the evaluation.
    b. Do not modify the given libraries, type declarations, prototypes, menus or comments in the supplied .cpp template.
    c. Complete every required function in the comment blocks and use the exact names and parameters supplied in the question paper and template.
    d. The supplied input file is **charging_sessions.txt.**

C++ FILE CODE EXPLANATION

The supplied C++ file sets up a menu-driven **EV Charging Session Analysis**. It imports records from **charging_sessions.txt**, stores them in a dynamically allocated array, and exports the required report to **charging_peak_summary.txt**.

The system includes functionalities for:

- **Load:** Charging sessions from a text file.
- **Calculate:** Tiered charge costs.
- **Identify:** Peak-energy sessions with a template.
- **Export:** A peak-session summary.

The program uses a typedef structure named ChargingSession with direct data members, a static reference record accessed with the dot operator, and a dynamic array accessed through a pointer and arrow operator.

The required type declarations and functions below must be implemented in the appropriate comment blocks in the supplied .cpp template file.

Use the exact function names and parameters from the supplied template.

STRUCTURE DEFINITION EXPLANATION

The ChargingSession structure and supporting types define the properties and operations required by this engineering system:

- **Structure and type requirements:**
    - **stationID** (int) and startHour (int): session identity.
    - **sessionCode** (char): C for charging or F for faulted.
    - **recordedAmount** (float): direct energy reading or recorded fault value.
    - **tariffBand** (char): low or peak tariff selector.
    - **sessions** pointer and arraySize: dynamic session data.

- **Required functions:**
    - **InitialiseChargingSession()**: initialise one record.
    - **TextFileLineCount()**: validate and count file records.
    - **ReadFileAndPopulate()**: create/reload dynamic data.
    - **CalculateChargeCost()**: perform the scenario calculation.
    - **DisplayTariffName()**: display scenario information.
    - **FindPeak()**: complete the rotated Unit 1 technique.
    - Report and cleanup functions: export results and release dynamic memory.

FUNCTION EXPLANATIONS

1. InitialiseChargingSession Function

void ChargingSession::InitialiseChargingSession(int, int, char, float, char)

Purpose: Initialises a charging or faulted session using direct structure members.
Parameters: identifier, hour, type, value and tariff band.
Returns: No value.

---

2. TextFileLineCount Function

int TextFileLineCount(string fileName)

Purpose: Checks the supplied file and counts non-empty session records.
Parameters: fileName: input text-file name.
Returns: Number of valid records, or zero.

---

3. ReadFileAndPopulate Function

void ReadFileAndPopulate(string, ChargingSession **, int *)

Purpose: Replaces existing records with a dynamically allocated, arrow-populated array.
Parameters: file name, session-array pointer and size pointer.
Returns: No value; updates the caller data.

---

4. CalculateChargeCost Function

float CalculateChargeCost(ChargingSession session)

Purpose: Calculates low/peak tariff cost and returns zero for a faulted session.
Parameters: session: one charging-session structure.
Returns: Charge cost.

---

5. DisplayTariffName Function

void DisplayTariffName()

Purpose: Uses a local tariffName and ::tariffName to display the global label.
Parameters: None.
Returns: No value; outputs to the console.

---

6. FindPeak Function

template <typename T> T FindPeak(T first, T second)

Purpose: Returns the higher same-type value to identify maximum session energy.
Parameters: first and second values of the same type.
Returns: The higher value.

---

7. DisplayPeakSessions Function

void DisplayPeakSessions(ChargingSession *, int)

Purpose: Displays the peak-energy station and its calculated charge cost.
Parameters: array pointer and number of sessions.
Returns: No value; outputs to the console.

---

8. Summary Export and Cleanup Functions

WritePeakSummary(...) and DeleteSessionArray(...)

Purpose: Exports peak-band sessions, then releases and resets dynamic session data.
Parameters: Use the exact pointers and size declared in the template.
Returns: No value; writes the summary and cleans up.

PRINT SCREENS

Main menu:

```
EV Charging Session Analysis
ES216BB
Evaluation 1

1. Load sessions
2. Display peak sessions
3. Display tariff label
4. Export peak summary
5. Exit
```

No data loaded:

```
EV Charging Session Analysis
ES216BB
Evaluation 1

1. Load sessions
2. Display peak sessions
3. Display tariff label
4. Export peak summary
5. Exit
Choice: 2

No sessions are loaded.

Press any key to continue...
```

Load charging sessions:

```
EV Charging Session Analysis
ES216BB
Evaluation 1

1. Load sessions
2. Display peak sessions
3. Display tariff label
4. Export peak summary
5. Exit
Choice: 1

Input file: charging_sessions.txt
5 charging sessions loaded.

Press any key to continue...
```

Display session costs:

```
EV Charging Session Analysis
ES216BB
Evaluation 1

1. Load sessions
2. Display peak sessions
3. Display tariff label
4. Export peak summary
5. Exit
Choice: 2

Peak energy: 45.0 kWh
Station 402, cost R153.0

Press any key to continue...
```

Identify peak session:

```
EV Charging Session Analysis
ES216BB
Evaluation 1

1. Load sessions
2. Display peak sessions
3. Display tariff label
4. Export peak summary
5. Exit
Choice: 3

EV Charging Tariff - Selected tariff display

Press any key to continue...
```

Export peak summary:

```
EV Charging Session Analysis
ES216BB
Evaluation 1

1. Load sessions
2. Display peak sessions
3. Display tariff label
4. Export peak summary
5. Exit
Choice: 4

Peak summary written to charging_peak_summary.txt


Press any key to continue...
```

Exit application:

```
EV Charging Session Analysis
ES216BB
Evaluation 1

1. Load sessions
2. Display peak sessions
3. Display tariff label
4. Export peak summary
5. Exit
Choice: 5

Session analyser closed.
```

Text File Input:

charging_sessions.txt - Notepad

```
401 8 C 28.5 L
402 18 C 45.0 P
403 19 F 31 P
404 11 C 12.0 L
405 20 C 38.0 P
```

Ln 1, Col 1 | 100% | Unix (LF) | UTF-8

Text File Output:

charging_peak_summary.txt - Notepad

```
EV CHARGING PEAK SESSION SUMMARY
Station 402: R153.00
Station 405: R129.20
```

Ln 4, Col 1 | 100% | Windows (CRLF) | UTF-8

ANNEXURE A – MARK ALLOCATION

Note: Score range is 0 - 4: 0-none, 1-poor, 2-average, 3-good, 4-excellent

TEST RUBRIC	SCORE [0-4]	WEIGHT [%]
C++ CODE EVALUATION		**50**
0. Structures, types, static and dynamic access		**5**
1. Function1		**5**
2. Function2		**5**
3. Function3		**5**
4. Function4		**5**
5. Function5		**5**
6. Function6		**5**
7. Function7		**5**
8. Function8		**5**
9. Compile and runtime stability		**5**
TOTAL		**50**

Graduate Attribute	GA Number	GA Score [0-5]
Application of scientific and engineering knowledge	GA2	0,1,4
Engineering methods, skills, tools, including information technology	GA5	1,2,3,5,6
Impact of Engineering Activity	GA7	4,5,6
Engineering Professionalism	GA10	7,8,9

ANNEXURE B – INFORMATION SHEET

Data types: void, char, short, int, float, double

Data Type modifiers: const, auto, static, unsigned, signed

Arithmetic operators: * / % + -

Relational operators: < <= > >= == !=

Assignment operator: = += -= *= /= %= &= ^= |= <<= >>=

Logic operators: && || !

Bitwise logic operators: & | ^ ~ << >>

Pointer operators: Derefernce: * Address: &

Control Structures:

IF Selection: if (condition) { … };

IF ELSE Selection: if (condition) { … } else { … };

WHILE Loop: while (condition) { … };

DO WHILE loop: do { … } while (condition);

FOR Loop: for (initial value of control variable; loop condition; increment of control variable) { … }

SWITCH Selection: switch (control variable){ case ‘value’: … ; break; default: … ; break; }

Functions: return_data_type function_name (parameters) { … };

Arrays:

One dimensional: data_type variable_name[size];

Two dimensional: data_type variable_name [x_size][y_size];

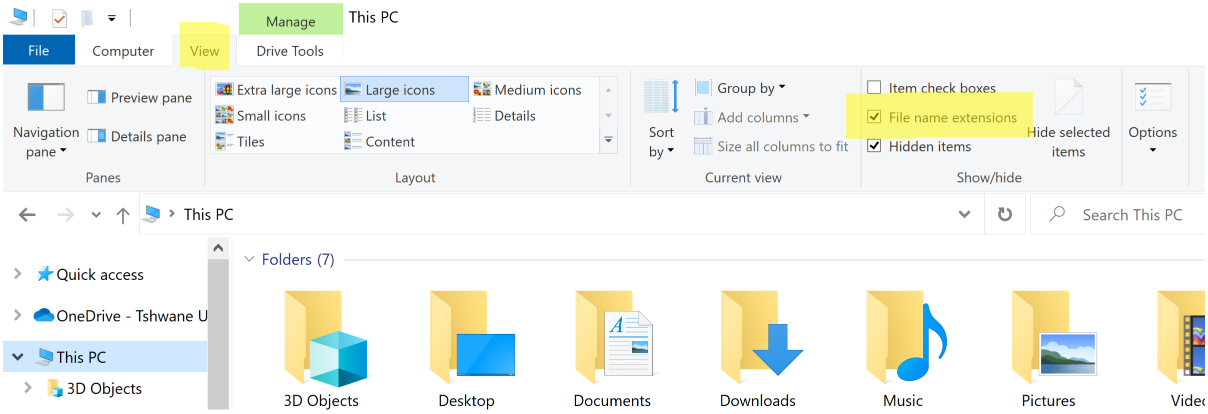
ANNEXURE C – ASCII TABLE

Dec	Hx	Oct	Char		Dec	Hx	Oct	Html	Chr	Dec	Hx	Oct	Html	Chr	Dec	Hx	Oct	Html	Chr
0	0	000	NUL	(null)	32	20	040		Space	64	40	100	@	@	96	60	140	`	`
1	1	001	SOH	(start of heading)	33	21	041	!	!	65	41	101	A	A	97	61	141	a	a
2	2	002	STX	(start of text)	34	22	042	"	"	66	42	102	B	B	98	62	142	b	b
3	3	003	ETX	(end of text)	35	23	043	#	#	67	43	103	C	C	99	63	143	c	c
4	4	004	EOT	(end of transmission)	36	24	044	$	$	68	44	104	D	D	100	64	144	d	d
5	5	005	ENQ	(enquiry)	37	25	045	%	%	69	45	105	E	E	101	65	145	e	e
6	6	006	ACK	(acknowledge)	38	26	046	&	&	70	46	106	F	F	102	66	146	f	f
7	7	007	BEL	(bell)	39	27	047	'	'	71	47	107	G	G	103	67	147	g	g
8	8	010	BS	(backspace)	40	28	050	(	(	72	48	110	H	H	104	68	150	h	h
9	9	011	TAB	(horizontal tab)	41	29	051	)	)	73	49	111	I	I	105	69	151	i	i
10	A	012	LF	(NL line feed, new line)	42	2A	052	*	*	74	4A	112	J	J	106	6A	152	j	j
11	B	013	VT	(vertical tab)	43	2B	053	+	+	75	4B	113	K	K	107	6B	153	k	k
12	C	014	FF	(NP form feed, new page)	44	2C	054	,	,	76	4C	114	L	L	108	6C	154	l	l
13	D	015	CR	(carriage return)	45	2D	055	-	-	77	4D	115	M	M	109	6D	155	m	m
14	E	016	SO	(shift out)	46	2E	056	.	.	78	4E	116	N	N	110	6E	156	n	n
15	F	017	SI	(shift in)	47	2F	057	/	/	79	4F	117	O	O	111	6F	157	o	o
16	10	020	DLE	(data link escape)	48	30	060	0	0	80	50	120	P	P	112	70	160	p	p
17	11	021	DC1	(device control 1)	49	31	061	1	1	81	51	121	Q	Q	113	71	161	q	q
18	12	022	DC2	(device control 2)	50	32	062	2	2	82	52	122	R	R	114	72	162	r	r
19	13	023	DC3	(device control 3)	51	33	063	3	3	83	53	123	S	S	115	73	163	s	s
20	14	024	DC4	(device control 4)	52	34	064	4	4	84	54	124	T	T	116	74	164	t	t
21	15	025	NAK	(negative acknowledge)	53	35	065	5	5	85	55	125	U	U	117	75	165	u	u
22	16	026	SYN	(synchronous idle)	54	36	066	6	6	86	56	126	V	V	118	76	166	v	v
23	17	027	ETB	(end of trans. block)	55	37	067	7	7	87	57	127	W	W	119	77	167	w	w
24	18	030	CAN	(cancel)	56	38	070	8	8	88	58	130	X	X	120	78	170	x	x
25	19	031	EM	(end of medium)	57	39	071	9	9	89	59	131	Y	Y	121	79	171	y	y
26	1A	032	SUB	(substitute)	58	3A	072	:	:	90	5A	132	Z	Z	122	7A	172	z	z
27	1B	033	ESC	(escape)	59	3B	073	;	;	91	5B	133	[	[	123	7B	173	{	{
28	1C	034	FS	(file separator)	60	3C	074	<	<	92	5C	134	\	\	124	7C	174			|
29	1D	035	GS	(group separator)	61	3D	075	=	=	93	5D	135	]	]	125	7D	175	}	}
30	1E	036	RS	(record separator)	62	3E	076	>	>	94	5E	136	^	^	126	7E	176	~	~
31	1F	037	US	(unit separator)	63	3F	077	?	?	95	5F	137	_	_	127	7F	177	&#127;	DEL

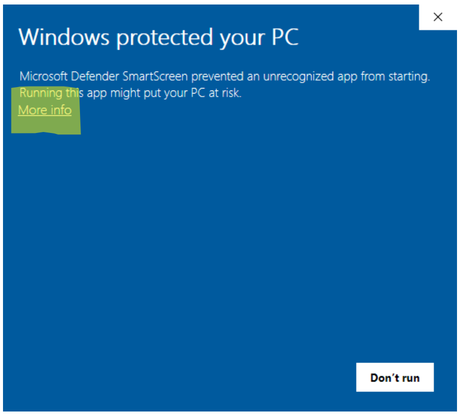
Source: www.LookupTables.com

ANNEXURE D – HOW TO RUN THE SHOWCASE FILE

1. Enable file extensions (see highlighted in yellow)



2. Change the name from **"Showcase.old"** to **"Showcase.exe"**
3. Run the **"ShowcaseEV.exe"** by double-clicking on the icon.
4. Windows may show the following. Click on **"More info"**



5. Click on **"Run anyway"**